Problem 1.

A.

Problem 2.

a) $v = \sqrt{\frac{2F\cos\theta L}{m}}$
b) $v = \sqrt{\frac{2(F(\cos\theta + \mu_k \sin\theta) - \mu_k mg)L}{m}}$

Problem 3.

a) $W_{\text{tension}} = m_2 g d$
$W_{\text{friction}} = -m_2 g d$
b) $W_{\text{tension}} = -m_2 g d$
$W_{\text{friction}} = m_2 g d$

Problem 4.

a) $W_{\text{friction}} = -401.7\text{ J}$
b) $v_3 = 5.89\text{ m/s}$
c) $d = 0.400\text{ m}$

Problem 5.

a) F.
b) D.
c) $v = \sqrt{\frac{1}{2}gh}$

Problem 6.

a) $P = -157\text{ W}$

Problem 7.

a) $v = 13.3\text{ m/s}$
b) $W = 770\text{ J}$

Problem 8.

a) $v = \sqrt{\frac{2(m_2 - \mu_k m_1)gl}{m_1 + m_2}}$